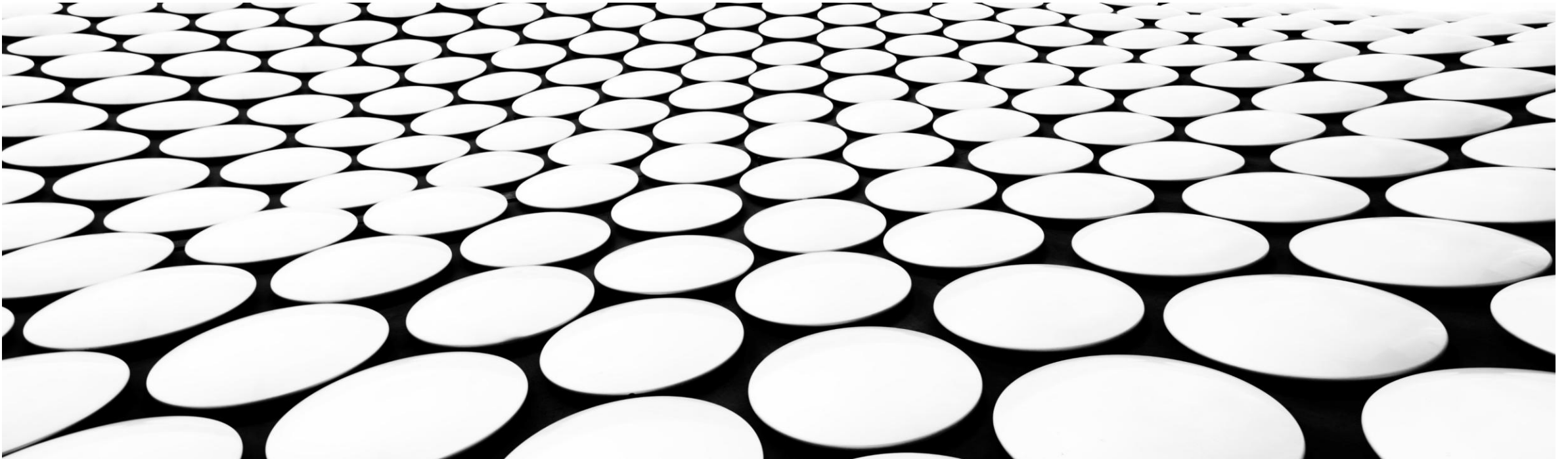
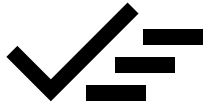


DECISION TREES

Underlying principles and examples





DECISION TREES

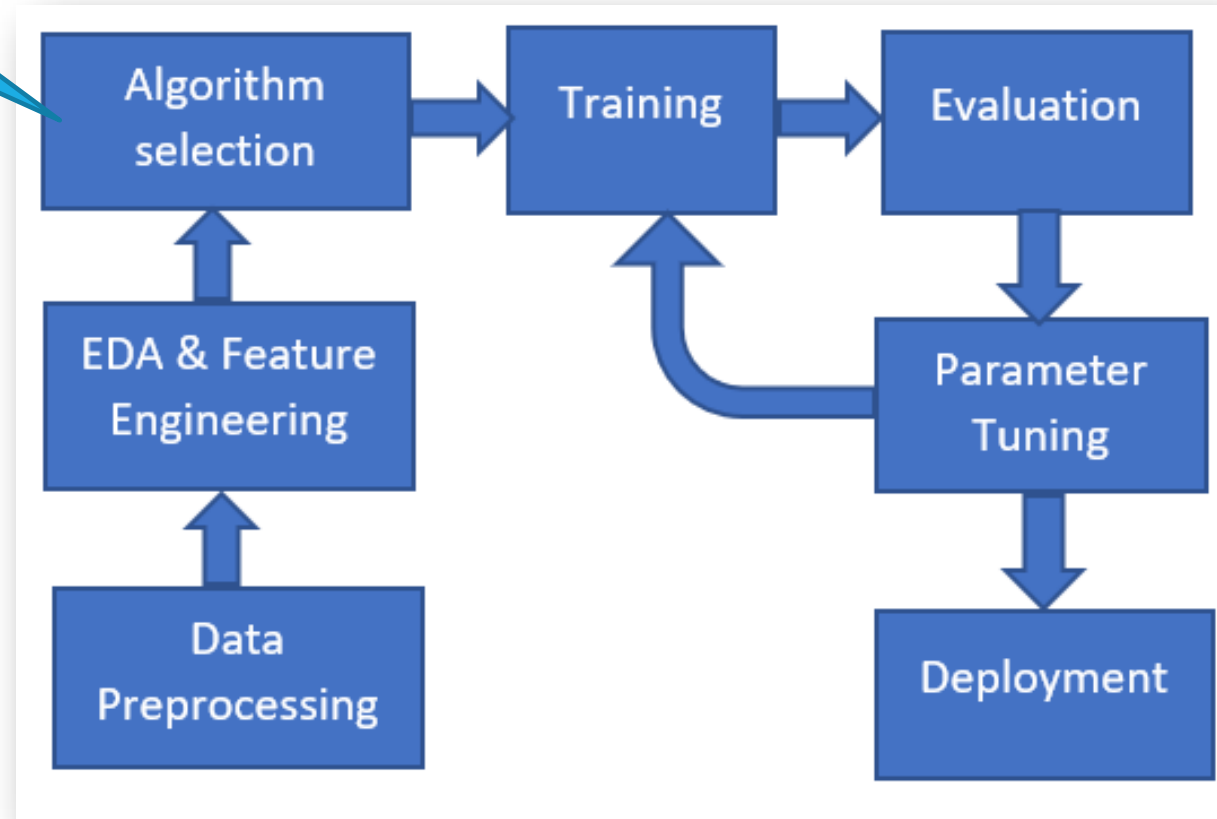
- Introduction to Decision Trees
- CART algorithm with numerical data

Look into Decision Trees

Classification
Naive Bayes
Decision Trees

Regression
Linear Regression

Supervised Machine Learning



INTRODUCTION

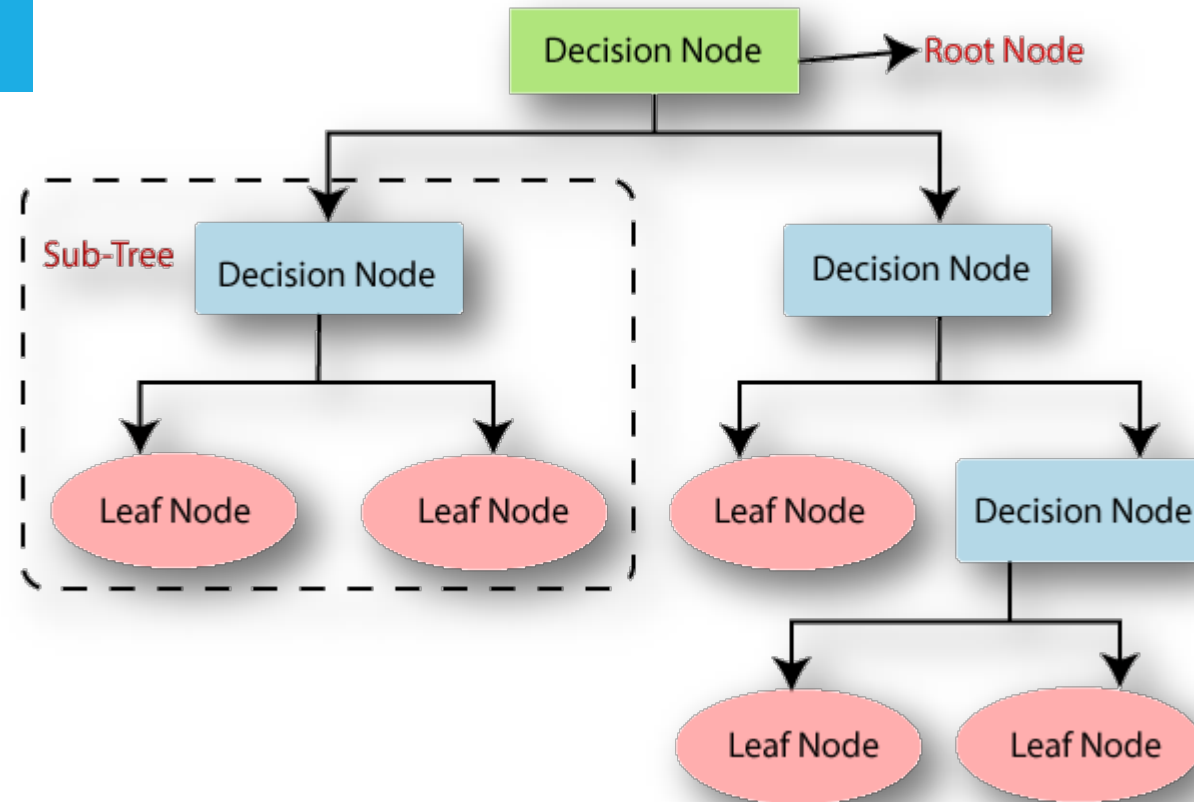
Definition

In machine learning, **decision trees** are one of the most intuitive and widely-used algorithms for classification and regression tasks.

They model decision-making processes in a tree-like structure where each internal node represents a feature (attribute), each branch represents a decision rule, and each leaf node represents the outcome.

Decision trees are easy to visualize and interpret, making them an excellent tool for explaining the logic behind predictions.

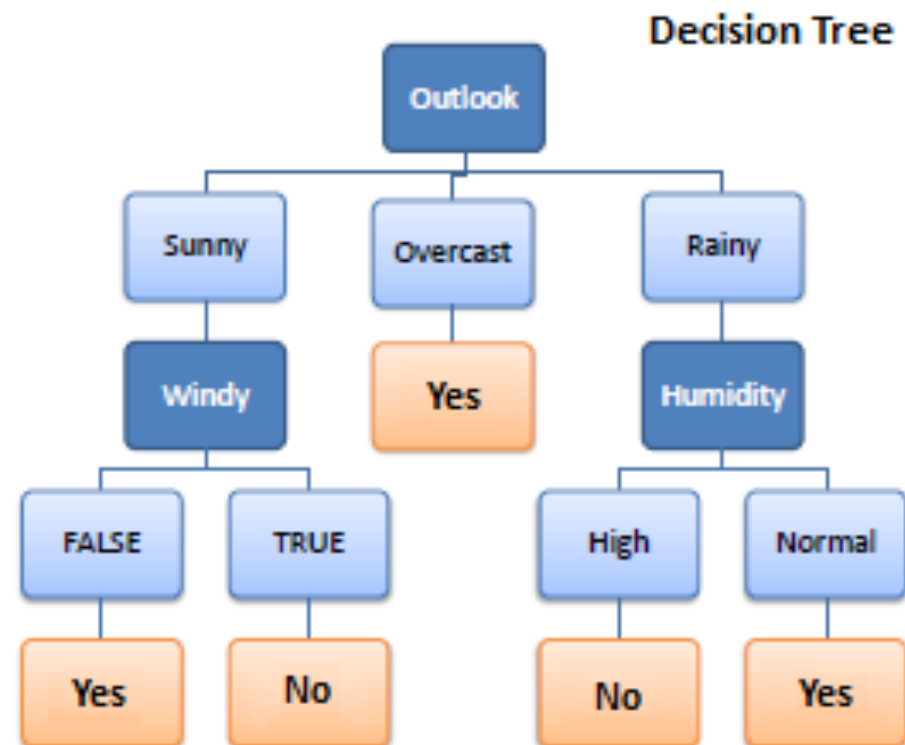
Terminology



Input/Output

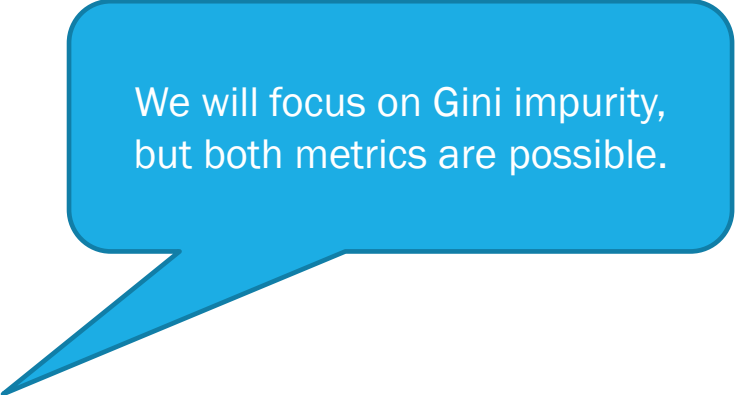
Decide on which attribute to split.

Predictors				Target
Outlook	Temp.	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No





Algorithm



We will focus on Gini impurity, but both metrics are possible.

How Decision Trees Work?

The process of creating a decision tree involves:

- 
- 1. Selecting the Best Attribute:** Using a metric like Gini impurity or information gain, the best attribute to split the data is selected.
 - 2. Splitting the Dataset:** The dataset is split into subsets based on the selected attribute.
 - 3. Repeating the Process:** The process is repeated recursively for each subset, creating a new internal node or leaf node until a stopping criterion is met (e.g., all instances in a node belong to the same class or a predefined depth is reached).

Classification and Regression Tree
– We only see Classification now

Numerical Features

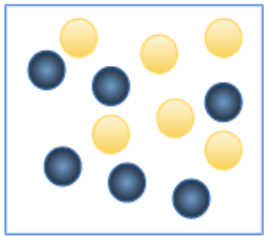
Example with CART (Gini Index)

Gini Index

Gini Impurity: Measures the frequency at which a randomly chosen element would be incorrectly classified. A lower Gini impurity indicates a purer node.

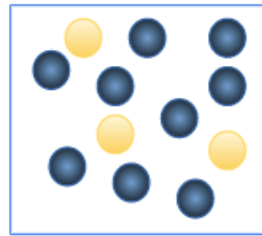
$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

Maximum Impurity (random)



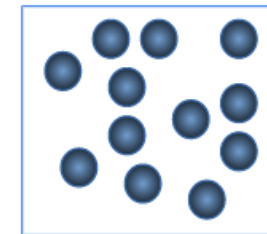
$$1 - (0.5^2 + 0.5^2) = 0.5$$

Average Impurity



$$1 - (0.75^2 + 0.25^2) = 0.375$$

Zero Impurity (unique class)



$$1 - (1^2 + 0^2) = 0$$

MACHINE LEARNING

Decision Tree Classifier, Explained: A Visual Guide with Code Examples for Beginners

A fresh look on our favorite upside-down tree

Samy Baladram
Aug 30, 2024 • 11 min readtowards
data science[Source](#)

Categorical

Sunny : 0/1

Overcast: 0/1

Rainy: 0/1

Windy: 0/1

One-hot encoding
(mutually exclusive)

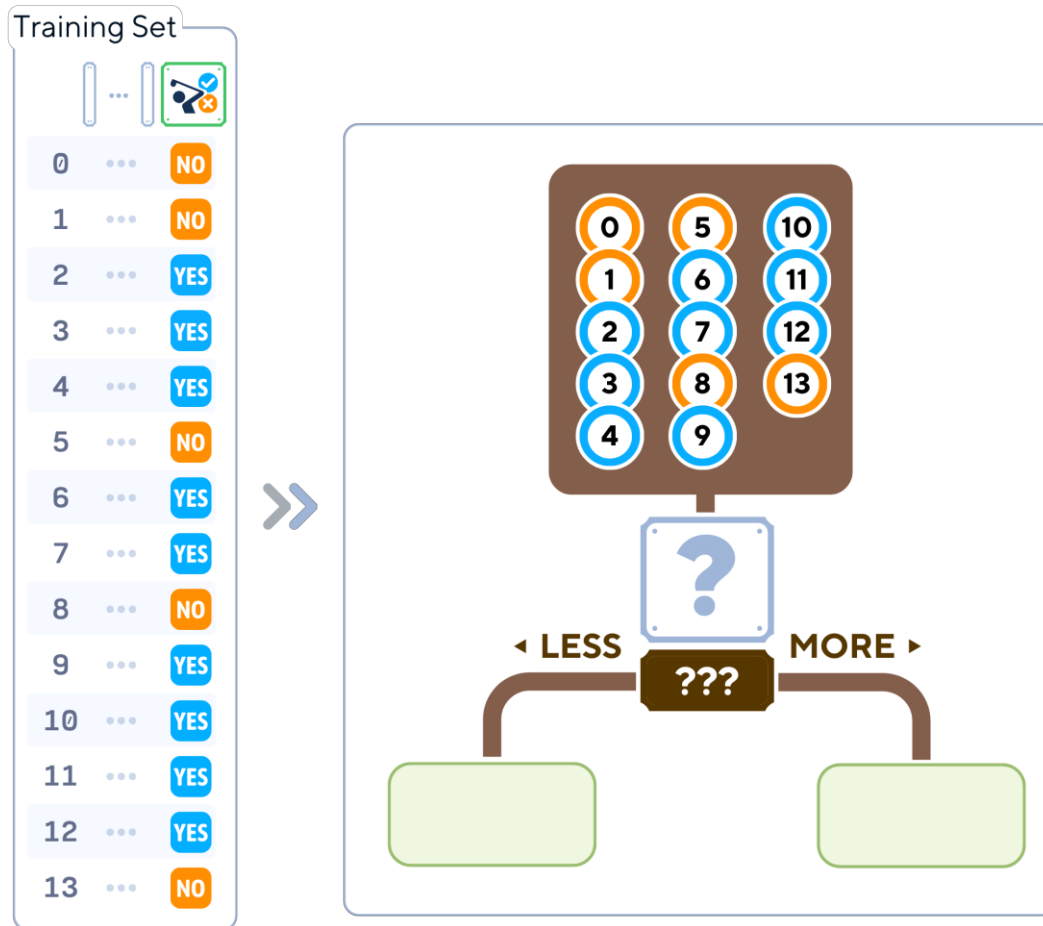
Numerical
Temperature
Humidity

Training Set



0	1	0	0	85	85	0	No
1	1	0	0	80	90	1	No
2	0	1	0	83	78	0	Yes
3	0	0	1	70	96	0	Yes
4	0	0	1	68	80	0	Yes
5	0	0	1	65	70	1	No
6	0	1	0	64	65	1	Yes
7	1	0	0	72	95	0	No
8	1	0	0	69	70	0	Yes
9	0	0	1	75	10	0	Yes
10	1	0	0	75	70	1	Yes
11	0	1	0	72	90	1	Yes
12	0	1	0	81	75	0	Yes
13	0	0	1	71	80	1	No

1. Start with all training samples in the root node.



Always the probability of
Yes/No (Play/Not play)

$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

Gini Index over the dataset

No = 5

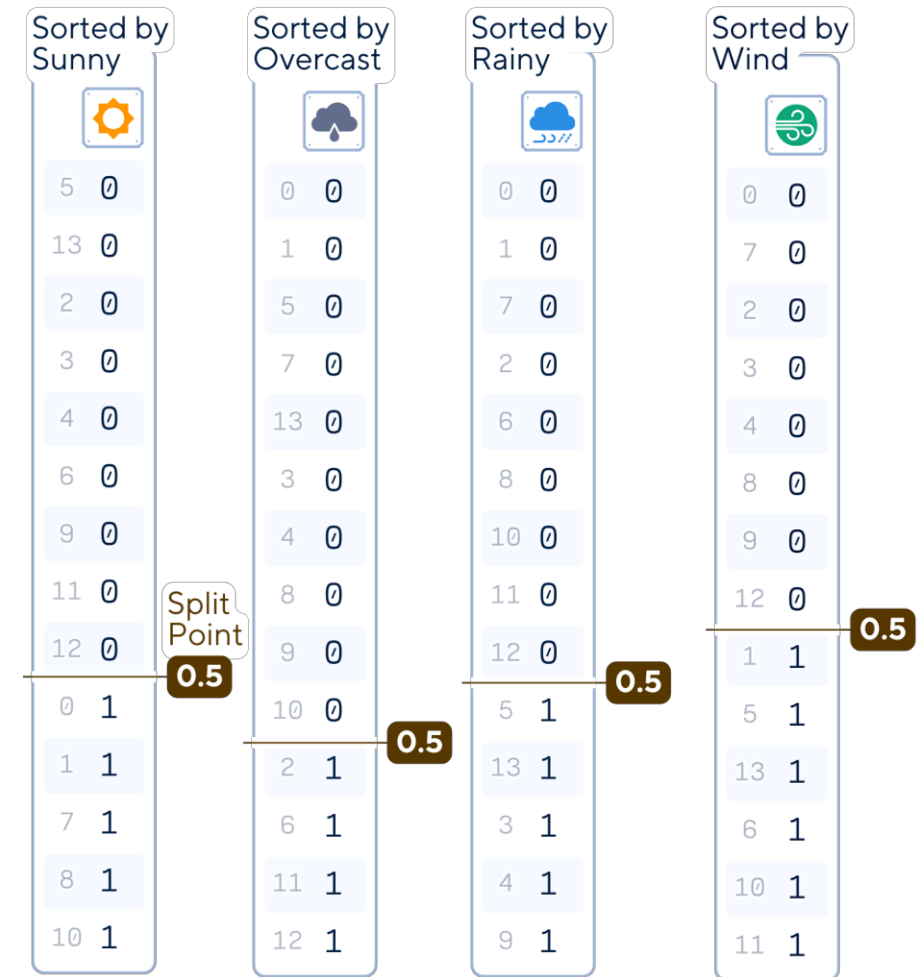
Yes = 9

$$1 - (5/14)^2 - (9/14)^2 = 0,46$$

2. For each feature:

- Sort the feature values.
- Consider all possible thresholds between adjacent values as potential split points.

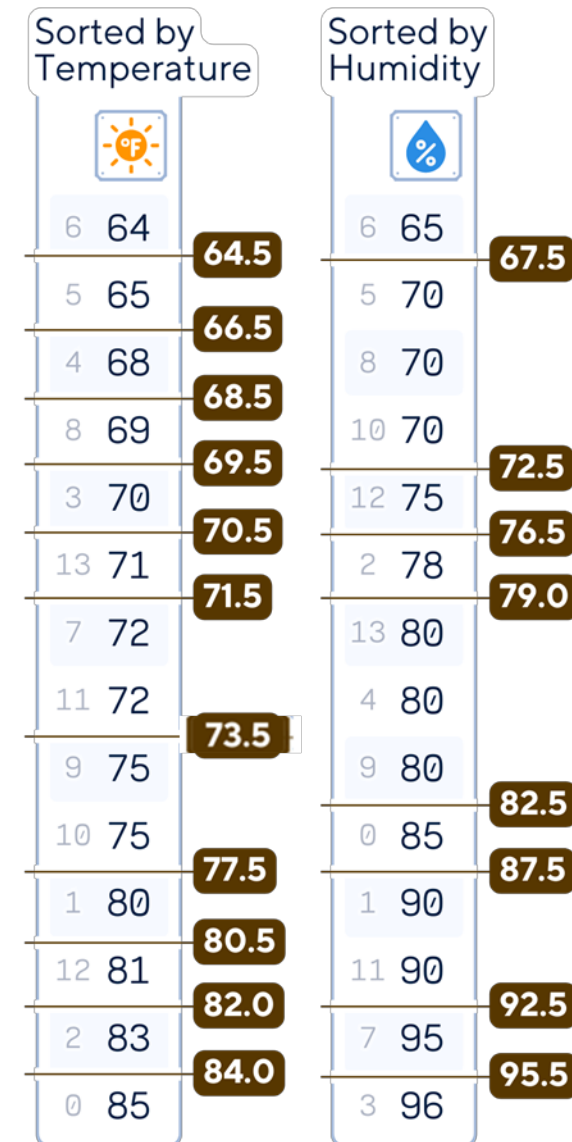
Single split point to check for the one-hot 0/1 attributes



2. For each feature:

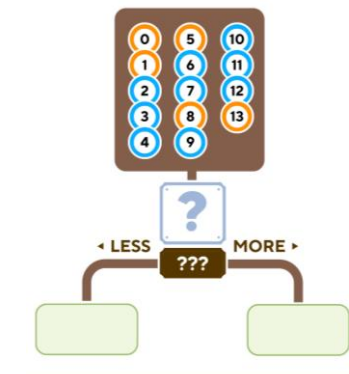
- Sort the feature values.
- Consider all possible thresholds between adjacent values as potential split points.

Lots of split points to check for numerical features...

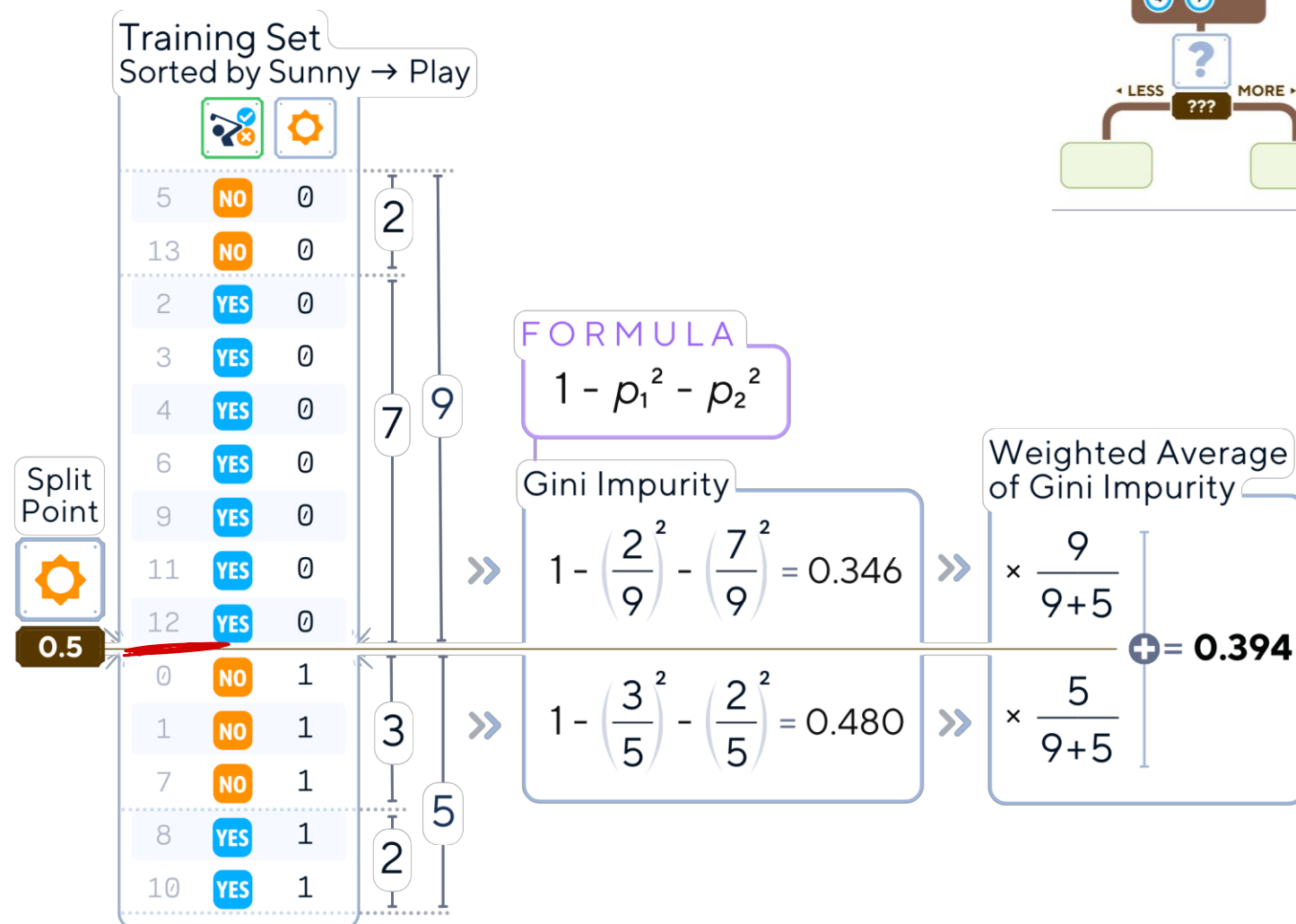


3. For each potential split point:

- Calculate the impurity (e.g, Gini impurity) of the current node.
- Calculate the weighted average of impurities.

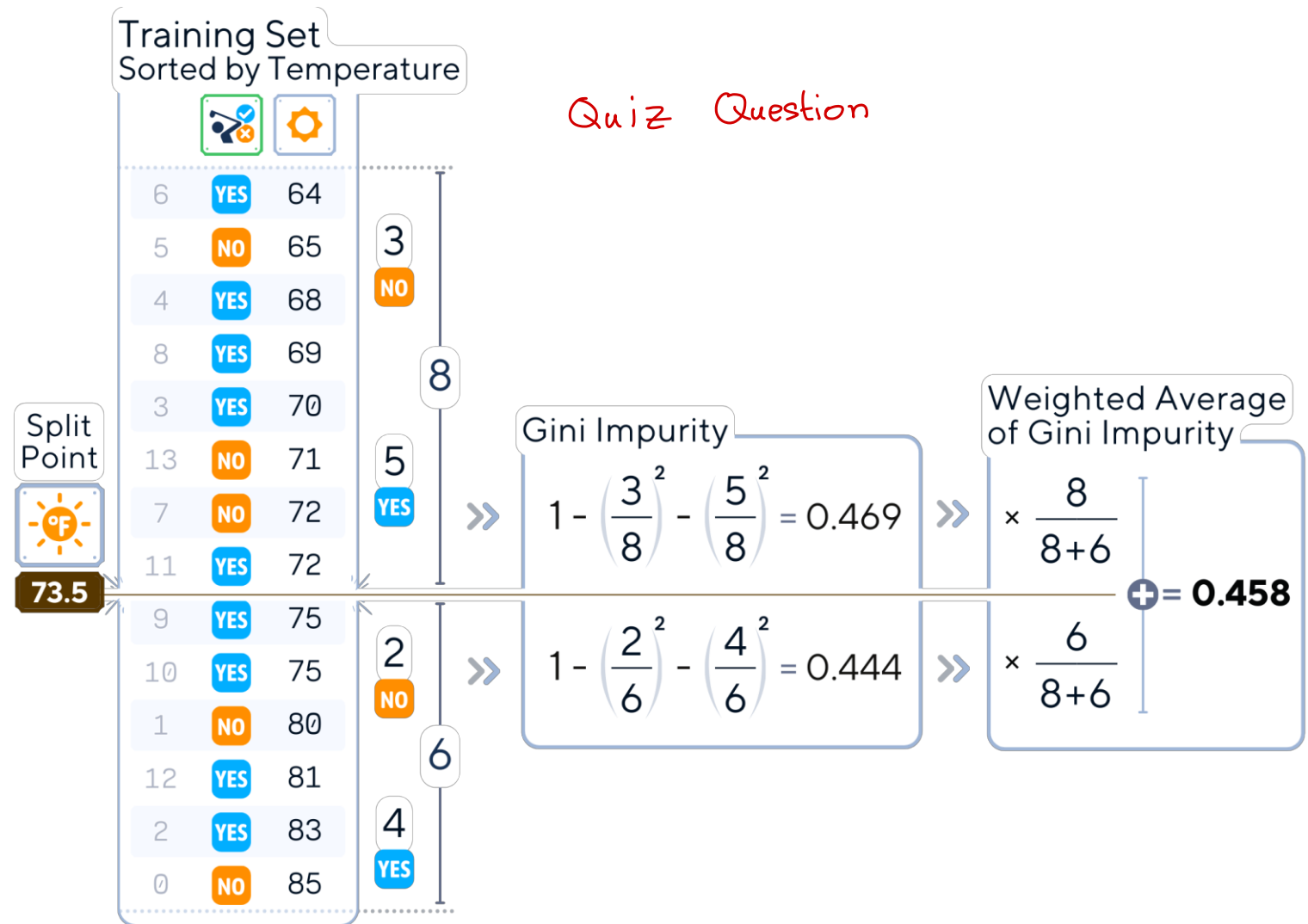


Single split point to test on Sunny feature



There are 11 split points to check... this is one of them

Quiz Question



4. After calculating all impurity for all features and split points, **choose the lowest one.**

Feature, Split Points & Weighted Average of Gini Impurity

 **0.5** 0.394

 **0.5** **0.357**

 **0.5** 0.457

 **64.5** 0.439

 **66.5** 0.452

 **68.5** 0.459

 **69.5** 0.450

 **70.5** 0.432

 **71.5** 0.458

 **73.5** 0.458

 **77.5** 0.443

 **80.5** 0.459

 **82.0** 0.452

 **84.0** 0.396

 **67.5** 0.439

 **72.5** 0.450

 **76.5** 0.432

 **79.0** 0.405

 **82.5** 0.394

 **87.5** 0.443

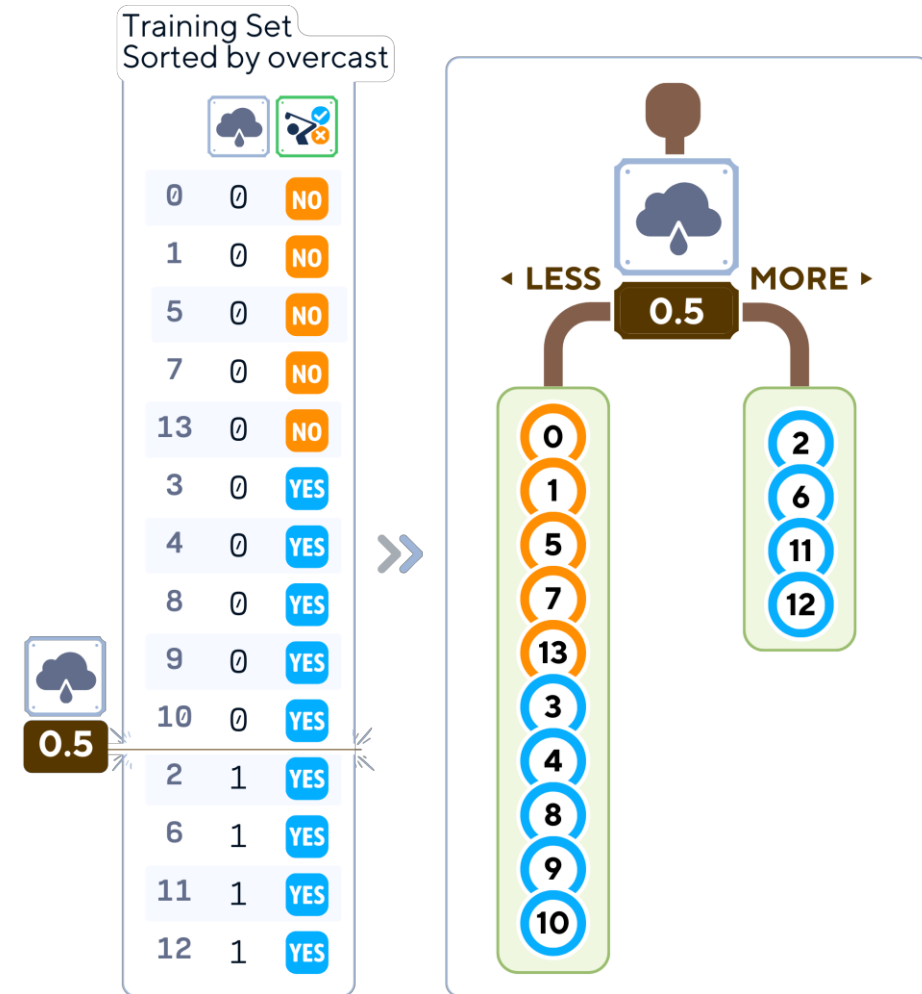
 **92.5** 0.452

 **95.5** 0.439

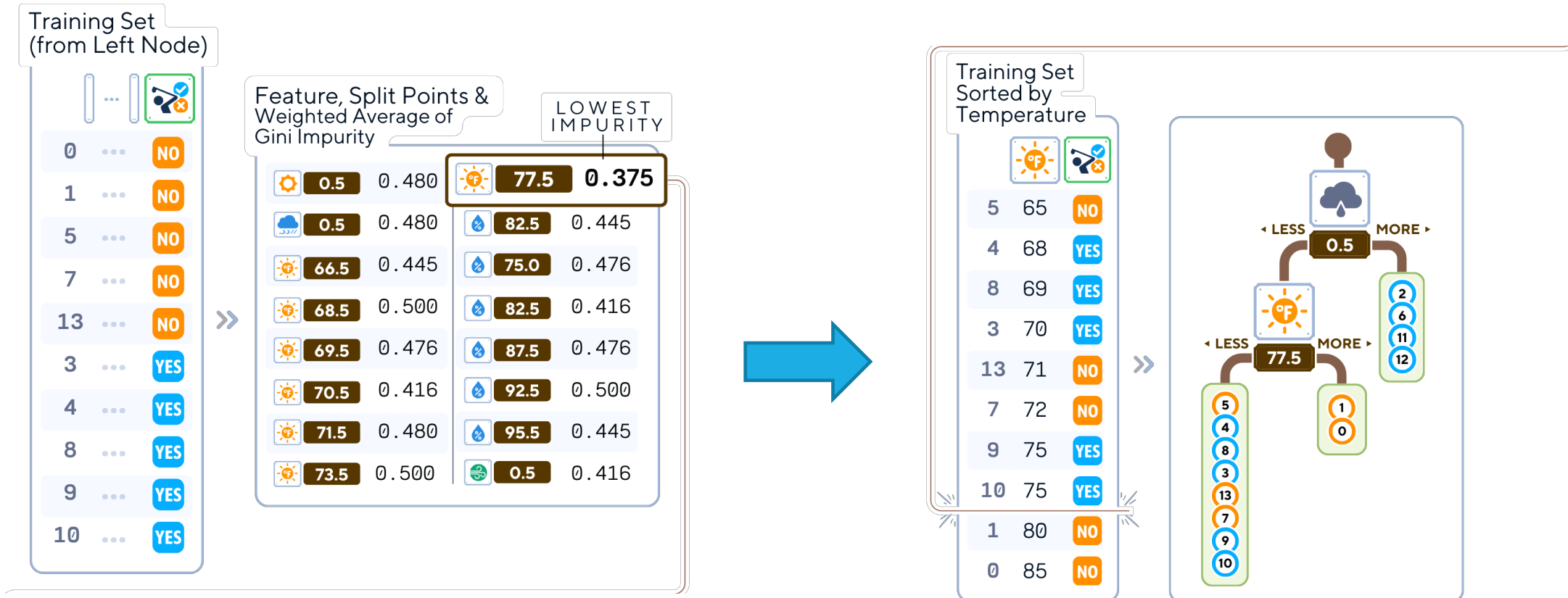
 **0.5** 0.429

5. **Create two child nodes** based on the chosen feature and split point:

- Left child: samples with feature value $\leq$ split point
- Right child: samples with feature value $>$ split point



6. Recursively repeat steps 2–5 for each child node.



Notice how each feature must be reconsidered...

Training Set
(from Left Node)

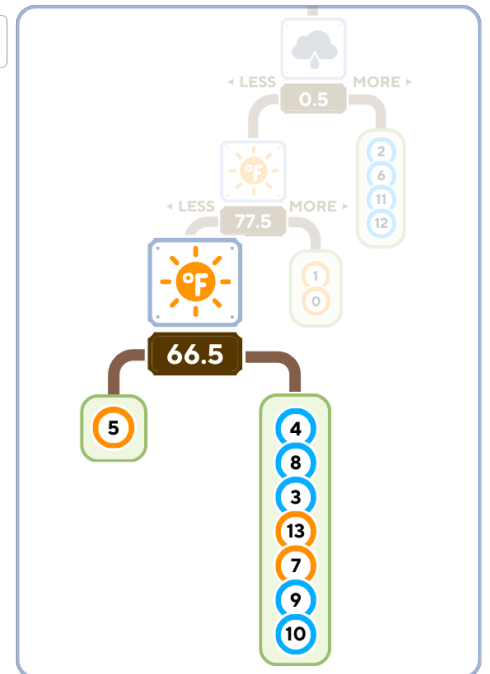
5	...	NO
4	...	YES
8	...	YES
3	...	YES
13	...	NO
7	...	NO
9	...	YES
10	...	YES

Feature, Split Points &
Weighted Average of
Gini Impurity

	0.5	0.467		71.5	0.467
	0.5	0.467		73.5	0.500
	66.5	0.357		75.0	0.467
	68.5	0.458		87.5	0.500
	69.5	0.467		95.5	0.429
	70.5	0.500		0.5	0.467

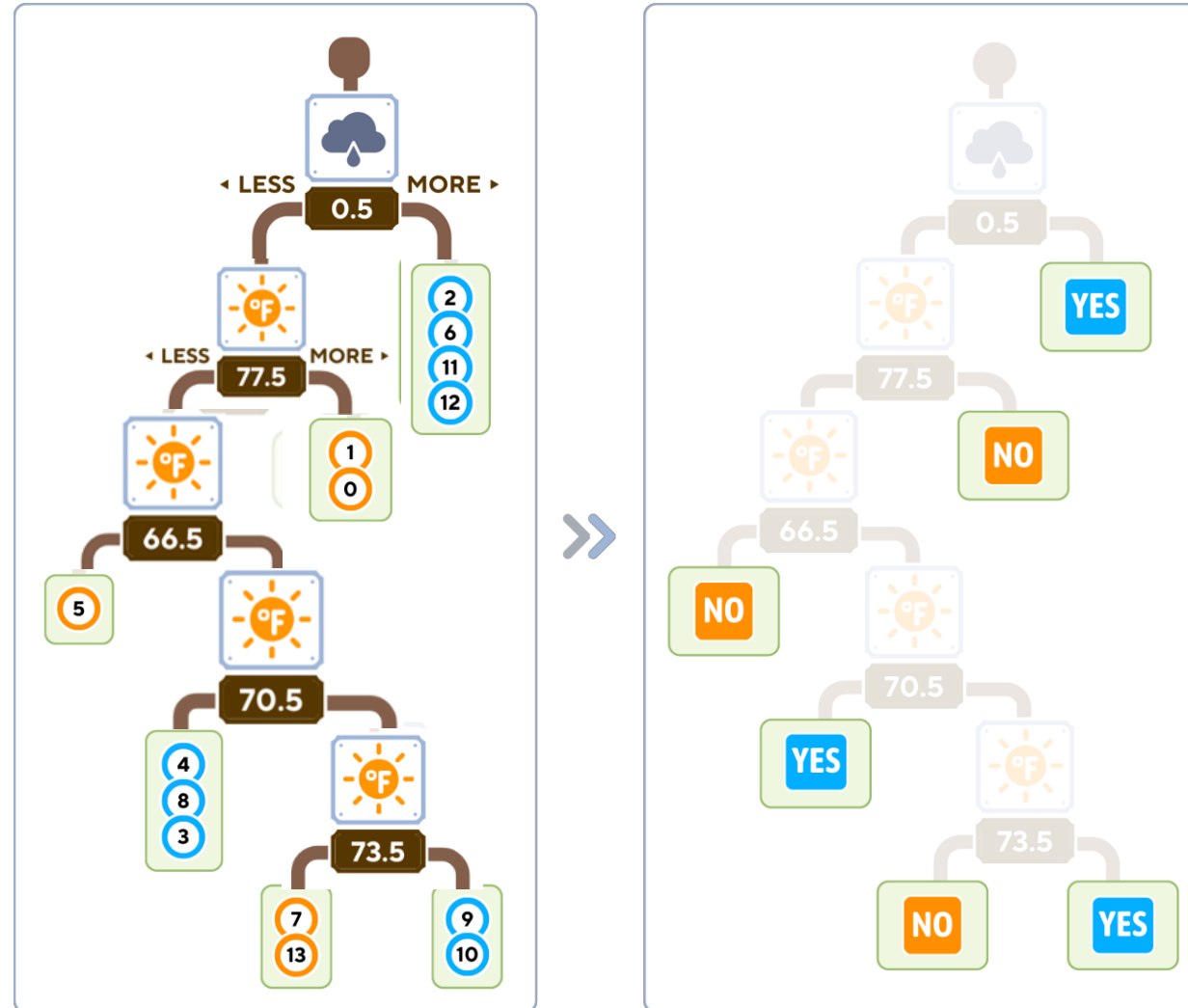
Training Set
Sorted by Temperature

5	65	NO
4	68	YES
8	69	YES
3	70	YES
13	71	NO
7	72	NO
9	75	YES
10	75	YES



Final Complete Tree

But do we really want
a complete tree?



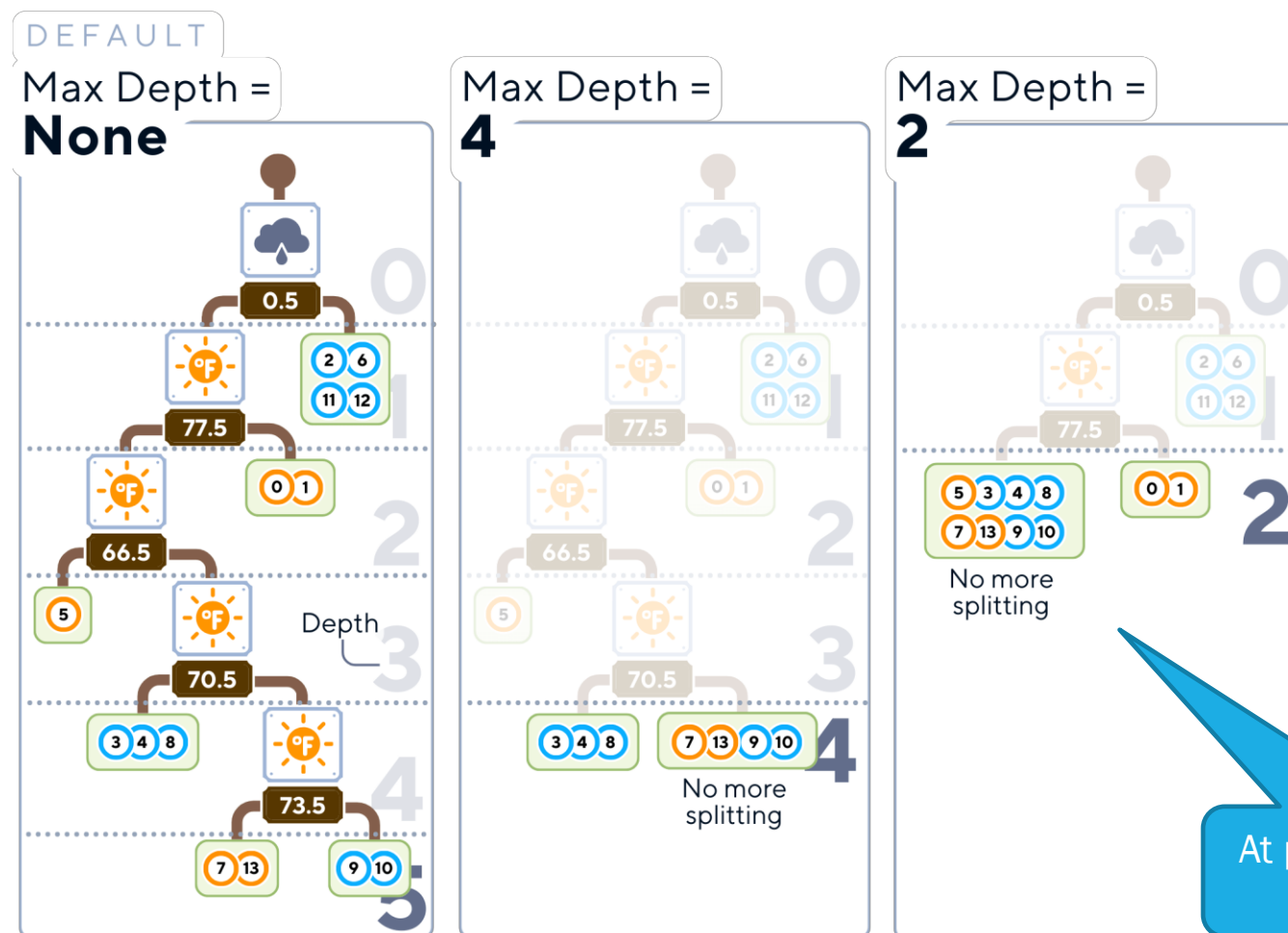
Stopping criteria

To prevent overfitting, decision trees use various stopping criteria:

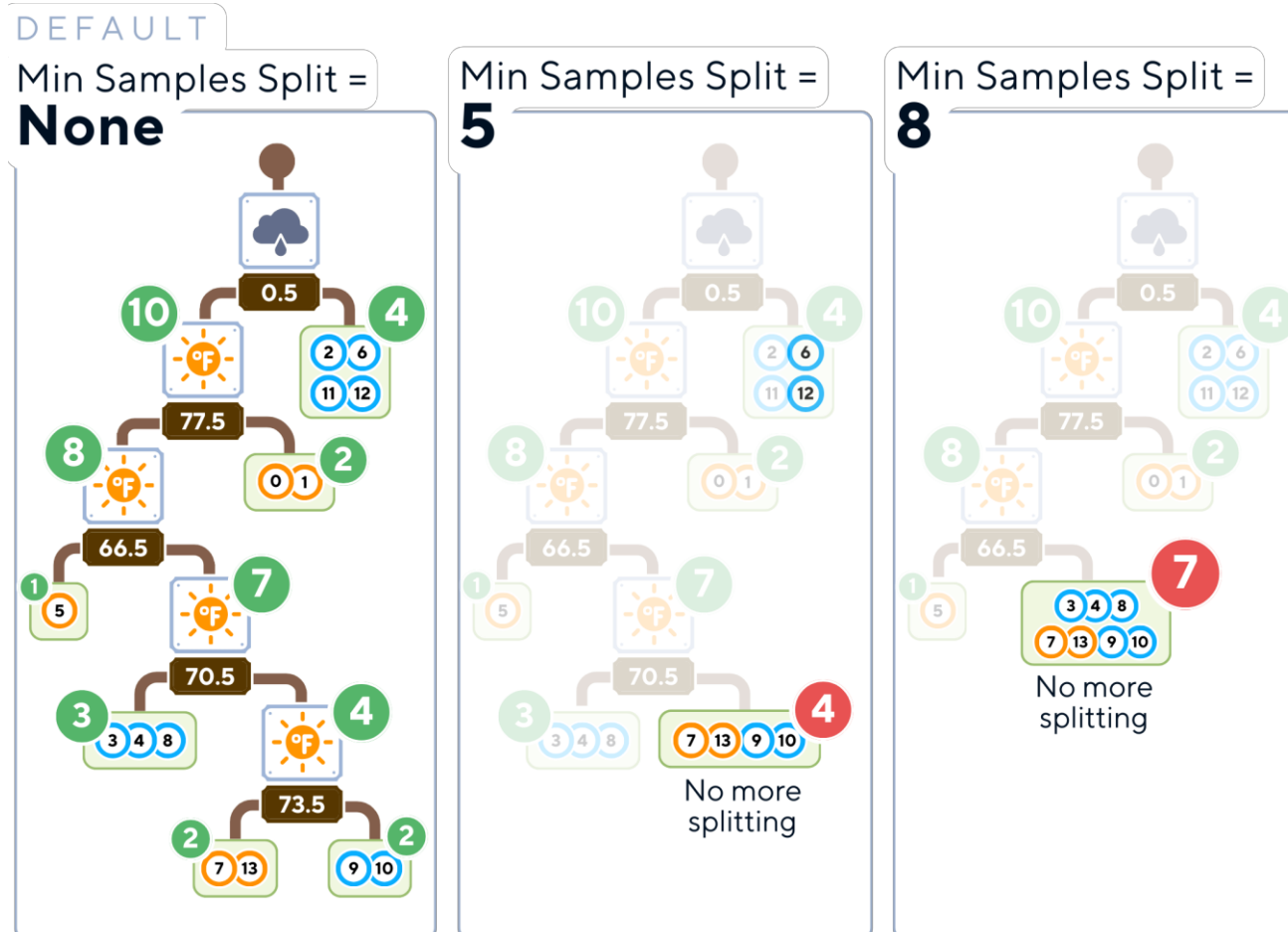
- **Maximum Depth:** Limits the depth of the tree.
- **Minimum Samples for a Split:** Ensures a node must have a minimum number of samples before splitting.
- **Minimum Samples for a Leaf:** Ensures each leaf node must have a minimum number of samples.

These will lead to impure leaves, but will allow generalization

Maximum Depth: Limits the depth of the tree.



Minimum Samples for a Split: Ensures a node must have a minimum number of samples before splitting.



At prediction time, use the mode

Advantages of Decision Trees

- 1. Interpretability:** Easy to understand and visualize the decision-making process.
- 2. No Feature Scaling:** Can handle both numerical and categorical data without normalization.
- 3. Handles Non-linear Relationships:** Can capture complex patterns in the data.

Interpretability

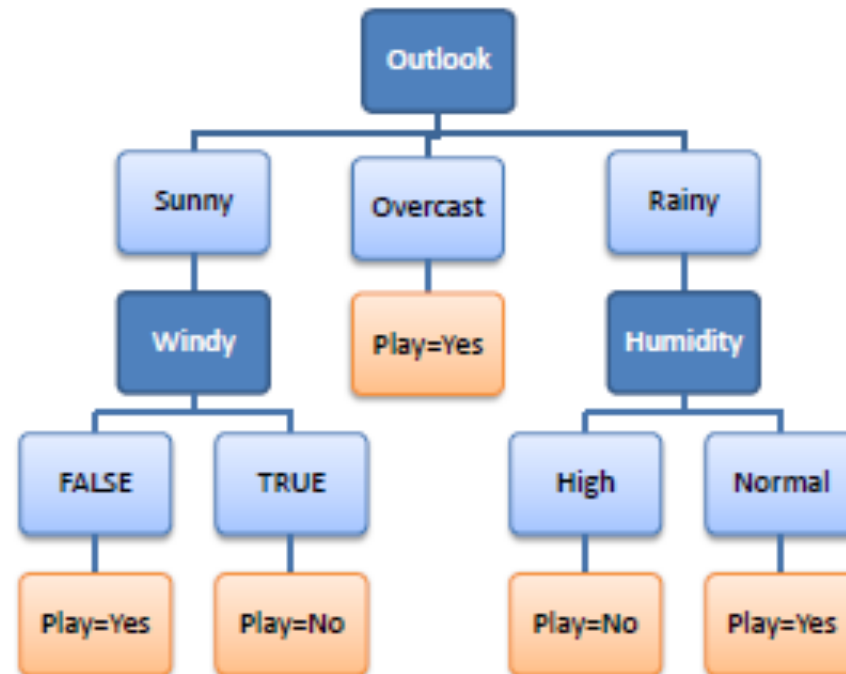
R_1 : IF (Outlook=Sunny) AND
(Windy=FALSE) THEN Play=Yes

R_2 : IF (Outlook=Sunny) AND
(Windy=TRUE) THEN Play=No

R_3 : IF (Outlook=Overcast) THEN
Play=Yes

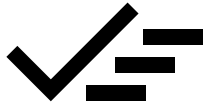
R_4 : IF (Outlook=Rainy) AND
(Humidity=High) THEN Play=No

R_5 : IF (Outlook=Rain) AND
(Humidity=Normal) THEN
Play=Yes



Disadvantages

- 1. Overfitting:** Prone to creating overly complex trees that don't generalize well, especially with small datasets.
- 2. Instability:** Small changes in the data can result in a completely different tree being generated.
- 3. Biased with Imbalanced Datasets:** Can be biased towards dominant classes.



DECISION TREES

- Introduction to Decision Trees
- CART algorithm with numerical data